AFF–815

Sub. Code
4BCE1C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science

PROGRAMMING IN C

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the character set used in C.
2. What are the rules to be followed to form an identifier?
3. Write the syntax for FOR statement.
4. What is the purpose of GOTO statement?
5. How will you initialize a character array?
6. What is recursion?
7. What is known as bit field?
8. How to use the pointer expression?
9. Write the usage of Macro substitution.
10. What is the purpose of getc ( ) and putc ( ) function?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the hierarchy of evaluation in an arithmetic expression.

Or

(b) Discuss the types of constants used in C.

12. (a) Write a C program to generate fibonacci series.

Or

(b) How to declare and initialize array? Explain with an example.

13. (a) Discuss the function with arguments and return values along with an example.

Or

(b) Write a C program to read a line of text from the terminal.

14. (a) How to declare the structure variable and discuss on how to access its members?

Or

(b) Explain how to pass pointers as function parameters with an example.

15. (a) How can you define and open a file?

Or

(b) Discuss on how to do the error handling during I/O operations.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe the different operators used in C. Give its hierarchy.

17. Explain two-dimensional arrays with a sample program.

18. Write a C program to read a string and count the number of characters in the string.

19. Define Pointers. How to declare and initialize the pointers? Explain it with an example.

20. Write a program in C to create an employee file and calculate the net pay of an employee.

AFF–816

Sub. Code
4BCESA1

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied — DIGITAL ELECTRONICS AND COMPUTER ARCHITECTURE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Draw the logic circuit of OR gate and its truth table.

2. What do you mean by sum-of-products expression? Give an example.

3. What are the various steps involved to design the combinational circuits?

4. What do you mean by FPLA?

5. What is a flip-flop? What is its function?

6. What does a Schmitt trigger do?

7. What is stack pointer?

8. Write the three types of CPU organizations.

9. What is called as address space and memory space?

10. Distinguish between Tightly coupled and Loosely coupled multiprocessor.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the basic laws of Boolean algebra? Discuss.

Or

(b) Draw K-map and simplify the Boolean expression

$$\overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + \overline{A}BCD + A\overline{B}\overline{C}D + A\overline{B}CD + AB\overline{C}D + ABC\overline{D}$$

12. (a) Explain a 3 to 8 line decoder circuit with truth table.

Or

(b) Draw the circuit of a decimal adder and explain its working.

13. (a) Explain J-K flipflop with neat diagram.

Or

(b) Draw the circuit diagram of a bi-directional shift register and explain its working.

14. (a) Evaluate the expression $X = (A + B) * (C + D)$ using zero, one, two and three address instructions.

Or

(b) Discuss on data manipulation instructions in a typical computer.

15. (a) Bring out the principle and constructional details of a magnetic disk.

Or

(b) What are the three types of mapping procedures in the Cache memory organization? Discuss.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Convert the following :

    (a) $(6747)_{10} = (\ )_{16}$

    (b) $(201)_{10} = (\ )_{8}$

    (c) $(704)_{8} = (\ )_{10}$

    (d) $(4C3F)_{16} = (\ )_{10}$

    (e) $(110111)_{2} = (\ )_{10}$.

17. With diagram explain the working of Half adder and full adder circuits.

18. Draw the circuit of a ripple up counter and using timing diagram explain its working.

19. What is the need for addressing modes? Explain the different types of addressing modes used in computers.

20. Explain the hardware organization of associative memory.

AFF–817

Sub. Code
4BCEA1

U.G. DEGREE EXAMINATION, NOVEMBER 2015.

***Allied* — OFFICE AUTOMATION**

(Offered by the Department of Computer Science)

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1½ = 15)

Answer **all** questions.

1. What is the purpose of formatting tool bar?

2. How to insert characters?

3. What do you mean by pattern matching?

4. How can you change page size and margins?

5. How can you change column width and row height?

6. List any three default file extensions.

7. Define Active cell.

8. Can you use mouse to change column width? Why?

9. Where will you use cut and paste commands in tables?

10. How can you sort records in a database?

Part B (5 × 3 = 15)

Answer **all** questions choosing either (a) or (b).

11. (a) What is overtyping? Explain.

Or

(b) How can you quickly open recently used files?

12. (a) Explain how to change spacing before and after a paragraph.

Or

(b) What is the use of page setup command in the file menu? Explain.

13. (a) Write short note on Word – Art in MS-Word.

Or

(b) Give an example for mail merge concept.

14. (a) Describe the organization of the Work Sheet Area.

Or

(b) Explain the concept of range with an example.

15. (a) How to insert a row and column in the work sheet?

Or

(b) What do you mean by print preview? Explain.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss on move and copy text.

17. Write about spell check and grammar.

18. How can you create tables? How to format it?

19. What are the charts that can be printed through Excel Sheet? Discuss them.

20. Describe how to sort and filter records.

AFF–818

Sub. Code
4BCE2C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science

PROGRAMMING IN C++ AND DATA STRUCTURES

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a variable?

2. Write the syntax "for" function prototype.

3. List out any two special characteristics of a static member variable.

4. What is a parameterized constructor?

5. What is a virtual base class?

6. What does "this" pointer point to?

7. Give examples for stacks.

8. What are the types of linked lists?

9. Define siblings.

10. Draw the diagram for a symmetric digraph.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the benefits of OOP?

Or

(b) What is identifier? What are the keywords used in C++?

12. (a) What is a friend function? What are the merits and demerits of using friend functions?

Or

(b) Write the rules for overloading operators.

13. (a) What is single inheritance? Explain with an example.

Or

(b) Write a C++ program to explain how to declare and define function pointers in C++.

14. (a) What are the operations on stack? Explain.

Or

(b) Describe how to insert a node in a DLL.

15. (a) How binary trees are represented?

Or

(b) What is graph? Explain the representation of graphs?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the formatted and unformatted console I/O operations available in C++.

17. What are destructors? Give an example to illustrate the destructor has been invoked implicitly by the complier.

18. Describe the concept of virtual functions.

19. Explain linked stacks and queues.

20. What is binary tree? Explain the different types of binary tree traversals.

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AFF–819

Sub. Code
4BCESA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied — MICRO PROCESSORS AND INTERFACING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is a bus? Why address is unidirectional?

2. What is wait state?

3. What is the use of REP prefix?

4. What are the various ADD instructions of 8086?

5. What is the function of $M/\overline{IO}$ in 8086?

6. What is USART?

7. What is keyboard encoder?

8. What is the function of a modem?

9. What is the memory addressing capacity of 8086?

10. Write the three operations modes of 80386 microprocessors.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What is a flag? Write the flags of 8086.

Or

(b) What is fetch and execute cycle? Explain.

12. (a) Explain the various string instructions of 8086.

Or

(b) Write an 8088 assembly language program to add two numbers and find the average.

13. (a) Discuss why an interrupt controller is required? Briefly explain the 8259 PIC.

Or

(b) Explain the working of 8257 programable DMA controller.

14. (a) Discuss on floppy disk sub system.

Or

(b) What are the different types of printers? Explain briefly.

15. (a) Explain the real address mode and PVAM mode of 80286.

Or

(b) Discuss the organization of registers in 80286. Explain how they are used.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With a block diagram explain the internal architecture of 8086 microprocessor.

17. List and explain the assembler directives of a 8086 assembler.

18. Draw the block diagram of 8255 programmable peripheral interface (PPI) and explain how the ports of PPI are programmed and the various modes of operation of the PPI.

19. What do you understand by CRT terminal? Discuss the working principle of a CRT display unit.

20. Give the block diagram of 80386 processor and explain its functional units.

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AFF–820

Sub. Code
4BCEA2

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied — PRINCIPLES OF INFORMATION TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A $(10 \times 1\frac{1}{2} = 15)$

Answer **all** questions.

1. What are the advantages of computers?

2. Write the major development of Computer Technology.

3. What is the use of spreadsheets?

4. Write the name of any three Application software.

5. Define Internet.

6. What is the role of local networks?

7. What is a file?

8. What is meant by compression?

9. What is a program?

10. What is the purpose of visual programming?

Part B (5 × 3 = 15)

Answer **all** questions.

11. (a) Discuss the resolution from analog to the digital age.

Or

(b) What are the generations of computers?

12. (a) Describe the basic features of user interface.

Or

(b) What are the advantages of word processors?

13. (a) How video conferencing is used?

Or

(b) Write a brief note on intranets.

14. (a) Give a short note on optical disks.

Or

(b) Explain the concept of key field.

15. (a) Discuss shortly on Internet programming.

Or

(b) Write a note on Management Information systems.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the six elements of computer and Communication system.

17. Write a detailed note on Database software.

18. Describe in detail about Communication channels.

19. Explain in detail about secondary storage devices.

20. Discuss the six phases of system analysis and design.

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AFF–821

Sub. Code
4BCE3C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science

JAVA PROGRAMMING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Define World Wide Web.

2. List out any two numeric data types available in Java.

3. What is the precedence of arithmetic operators?

4. Write the syntax for simple if statement.

5. Define Class.

6. What is an Array?

7. What is a Thread?

8. List out any two run-time errors.

9. Define Applet.

10. List out any two HTML tags.

Section B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) Discuss about the structure of Java Programs.

Or

(b) Write Short notes on Java Virtual Machine.

12. (a) Write about the nested if-else statement with an example.

Or

(b) Explain the "for" statement in Java with a suitable example.

13. (a) Explain how class members are accessed in Java.

Or

(b) Discuss about two dimensional arrays in Java.

14. (a) How will you create and access packages in Java? Explain with an example.

Or

(b) Explain the use of multiple catch statements in exception handling.

15. (a) Write about Applet Life Cycle.

Or

(b) Write a Java program to draw a polygon using graphics class.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the features of Java Language.

17. Discuss about the classification of Java operators.

18. Discuss about Inheritance in Java.

19. Explain Multi threaded programming.

20. Describe about graphics programming.

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AFF–822

Sub. Code
4BCESA3

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied — RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define 'Operations Research'.

2. Give any two features of OR.

3. Write the standard form of an LPP.

4. Give the condition for unbounded solution to the LPP.

5. Write the mathematical form of an Assignment problem.

6. When an Assignment problem is said to be unbalanced?

7. In a Transportation Problem, when a solution is said to be degenerate?

8. What are the methods available to find the initial solution in Transportation problem?

9. What is critical activity?

10. Give the formula to compute the variance of an activity.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe the methods that are available for deriving the solution to an OR model.

Or

(b) Write the limitations of OR.

12. (a) What are slack and surplus variables? Give examples.

Or

(b) Use graphical method to solve

Minimize $Z = -x_1 + 2x_2$

Subject to the constraints :

$$-x_1 + 3x_2 \leq 10$$
$$x_1 + x_2 \leq 6$$
$$x_1 - x_2 \leq 2$$
$$x_1, x_2 \geq 0.$$

13. (a) Solve the following assignment problem :

		Person			
		1	2	3	4
	A	10	12	19	11
Job	B	5	10	13	8
	C	12	14	13	11
	D	8	15	11	9

Or

(b) How can you formulate the Travelling Salesmen problem as an Assignment problem?

14. (a) Describe the Vogel's approximation method.

Or

(b) Determine an initial basic feasible solution to the following Transportation problem using North-west corner rule :

Destination

		D_1	D_2	D_3	D_4	Availability
	O_1	6	4	1	5	14
Origin	O_2	8	9	2	7	16
	O_3	4	3	6	2	5
Requirements		6	10	15	4	

15. (a) Write the rules for drawing network.

Or

(b) Construct the network for the following :

$A < D,E$; $B,D > F$; $C < G$; $B < H$; $F,G < I$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Classify the OR model and explain.

17. Solve the following LPP :

Maximize $Z = 3x_1 - x_2$

Subject to the constraints

$$2x_1 + x_2 \geq 2$$
$$x_1 + 3x_2 \leq 2$$
$$x_2 \leq 4$$
$$x_1,\ x_2 \geq 0.$$

18. Solve the following Assignment Problem :

		Job				
		I	II	III	IV	V
	1	2.5	5	1	6	1
	2	2	5	1.5	7	3
Machine	3	3	6.5	2	8	3
	4	3.5	7	2	9	4.5
	5	4	7	3	9	6
	6	6	9	5	10	6

19. Find the optimal solution to the Transportation problem :

					Availability
	6	1	9	3	70
	11	5	2	8	55
	10	12	4	7	90
Requirements	85	35	50	45	

20. A project schedule has the following characteristics :

Activity :	1-2	1-3	2-4	3-4	4-5	3-5
a :	2	9	5	2	6	8
m :	5	12	14	5	9	17
b :	14	15	17	8	12	20

Construct a PERT network and find the critical path.

AFF–823

Sub. Code
4BCEA3

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

Allied : PROGRAMMING IN C

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A $(10 \times 1\frac{1}{2} = 15)$

Answer **all** questions.

1. What is a type specifier?

2. What is a conditional expression?

3. What is the allowed data type for subscript?

4. How will you initialize a character array?

5. What for 'return' is used in C?

6. How will you access the individual members in a structure?

7. When pointers are useful?

8. Is the name of the array a pointer?

9. Give the syntax of fread( ) function.

10. What are the facilities offered by the preprocessor?

Part B (5 × 3 = 15)

Answer **all** questions.

11. (a) Describe the bit-wise operators available in C.

Or

(b) Explain the scanf( ) function with an example.

12. (a) How can you read strings from terminal?

Or

(b) How to dimensional arrays are declared?

13. (a) What is meant by call by value? Give an example.

Or

(b) Explain structures with in structure.

14. (a) Describe the operators exclusively used with pointers.

Or

(b) Describe the relationship between pointers and arrays.

15. (a) Name the functions used for random access files. Explain any one of them.

Or

(b) Describe the following conditional compilation :

(i) # if

(ii) # elif.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a program to print the prime numbers between two limits.

17. Describe the string handling functions available in C.

18. With a recursive function to evaluate $n!$ and write a program to evaluate $n\ C_r = \dfrac{n!}{r!\,(n-r)!}$.

19. Explain pointers to functions. Also distinguish between function returning pointers and pointers to functions.

20. Explain how can you handle error during I/O operations.

AFF–824

Sub. Code
4BCESA4

U.G. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

COMPUTER ORIENTED NUMERICAL METHODS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State the formula to find a root of the equation $f(x)=0$ which is between $x=a$ and $x=b$ of Regula-Falsi method.

2. What is the condition for converge of Gauss-Jacobi method of iteration?

3. What is meant by curve fitting?

4. Write down to normal equations to fit a quadratic curve by least square method.

5. If $f(x)=x^2-2x+2$ find $\Delta^2 f(x)$, taking the interval of differencing as 1.

6. State Newton's forward interpolation formula.

7. Write the formula for $\frac{dy}{dx}$ at $x=x_n$ using backward difference operator.

8. State Simpson's one-third rule.

9. What is the disadvantage in Taylor series method?

10. What is the error term in Milne's corrector formula?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Find a root of the equation $x^3 - 4x - 9 = 0$ correct to three decimal places using Bisection method.

Or

(b) Solve by Gauss-Elimination method.

$$2x + y + 4z = 12$$
$$8x - 3y + 2z = 20$$
$$4x + 11y - z = 33.$$

12. (a) Find a straight line fit of the form $y = a + bx$ by the method of group averages for the following data :

x	0	5	10	15	20	25
y	12	15	17	22	24	30

Or

(b) Fit a straight line to the following data, using the method of moments.

x	1	2	3	4
y	16	19	23	26

13. (a) Find the function whose first difference is $9x^2 + 11x + 5$.

Or

(b) Using Newton's formula, find the value of $f(1.5)$ from the following data :

x :	0	1	2	3	4
$f(x)$:	858.3	869.6	880.9	892.3	903.6

14. (a) Using the following data, find $f'(5)$.

x :	0	2	3	4	7	9
$f(x)$:	4	26	58	112	466	922

Or

(b) Dividing the range into 10 equal parts, find the approximate value of $\int_0^{\pi} \sin x dx$ by Trapezoidal rule.

15. (a) Find by Taylor's series method, the value of y at $x = 0.1$ and $x = 0.2$, to four decimal places from $y' = x^2 y - 1$, $y(0) = 1$.

Or

(b) Given :
$y' = xy + y^2, y(0) = 1, y(0.1) = 1.1169, y(0.2) = 1.2773$, $y(0.3) = 1.5049$, evaluate $y(0.4)$ by using Milne's method.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find, by Newton-Raphson method, the real root of $3x - \cos x - 1 = 0$.

17. Find the numerically largest eigen value and the corresponding eigen vector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

18. Use Lagrange's formula to fit a polynomial to the data :

x	–1	0	2	3
y	–8	3	1	12

19. Using Romberg's method to compute $\int_0^1 \frac{dx}{1+x^2}$ by taking $h = 0.5$ and 0.25.

20. Apply Runge-Kutta method to find approximate value of y for $x = 0.2$ in steps of 0.1 if $y' = x + y^2$ given that $y = 1$ when $x = 0$.